The Hidden Environmental Cost of Email Communication: A Green Computing Approach to Reducing Digital Carbon Footprint

Md. Ibrahim Zihad, Md. Foysal Hossain Khan, Shohada Rahman, Afia Anjum, Mahfuza Lubna Nuvia, and Prof. Dr. Md. Motaharul Islam

Department of Computer Science and Engineering,
United International University, Dhaka, Bangladesh
mzihad221257@bscse.uiu.ac.bd; mkhan221045@bscse.uiu.ac.bd;
srahman223628@bscse.uiu.ac.bd; aanjum223386@bscse.uiu.ac.bd;
mnuvia2230956@bscse.uiu.ac.bd; motaharul@cse.uiu.ac.bd

Abstract. Email communication is one of the most widely used digital communication methods in modern society. However, the environmental impact of excessive email usage is often overlooked. Every email activity including sending, receiving, storing, and forwarding, consumes energy through network infrastructure, data centers, and user devices. This energy consumption contributes to carbon emissions and increases the overall digital carbon footprint. The large number of unnecessary emails, spam messages, and large attachments increases data storage requirements and network traffic, leading to higher electricity consumption in data centers. The main problem addressed in this study is the hidden environmental pollution caused by inefficient email management. To address this issue, this paper proposes a Green Computing-based intelligent email management approach using machine learning models for efficient email management and raising mass awareness. We are using AI models like Random Forest to filter unwanted or spam emails, Decision Tree to categorize emails into important and non-important messages, and Linear Regression to estimate the carbon footprint generated by email activities. The effectiveness of this approach is evaluated by analyzing reductions in spam emails, storage usage, and estimated carbon emissions. By combining machine learning with green computing practices and promoting responsible email usage, this study demonstrates how organizations and individuals can reduce the digital carbon footprint associated with email communication and support environmental sustainability.

Keywords: Green Computing · Digital Carbon Footprint · Email Management · Machine Learning · Spam Detection · Carbon Emission Estimation

1 Introduction

The rapid advancement of digital technologies has transformed communication systems, making email one of the most widely used and essential tools in both

personal and professional environments. Billions of emails are exchanged daily across the globe, supporting business operations, academic activities, and social interactions [15,16]. Despite its convenience and efficiency, the environmental impact of email communication remains largely overlooked.

Every email sent, received, stored, or forwarded requires energy consumption across multiple layers of digital infrastructure, including user devices, communication networks, and data centers. These processes contribute to carbon emissions, forming a part of the broader digital carbon footprint [5,14,23]. Data centers, in particular, consume significant amounts of electricity for data storage, processing, and cooling, further intensifying the environmental impact of large-scale digital communication [9,18]. Recent research shows that optimizing data center energy consumption through green computing approaches can significantly reduce carbon footprint [18].

Although the carbon emission of a single email may appear negligible, its cumulative effect is substantial due to the enormous volume of global email traffic. Research indicates that standard emails produce small amounts of CO_2, whereas emails with large attachments generate significantly higher emissions. Additionally, the widespread presence of spam and unnecessary emails increases storage demand and network traffic, resulting in avoidable energy consumption and environmental degradation [4,17,19,20]. Advanced spam filtering and email classification techniques using machine learning have shown promise in reducing unnecessary email traffic [19,20].

Existing studies primarily focus on estimating carbon emissions and raising awareness about sustainable digital practices. However, they often lack practical, automated solutions to effectively manage email communication and reduce unnecessary digital activities. Most approaches emphasize behavioral changes rather than leveraging advanced technologies such as machine learning for optimization and decision-making [8,21]. The growing impact of digital technologies on the environment, particularly in data-intensive sectors, demands intelligent automation and monitoring systems [22,23].

To address this gap, this research adopts a green computing approach to develop an intelligent email management system aimed at reducing the digital carbon footprint. By integrating machine learning techniques such as Random Forest, Decision Tree, and Linear Regression, the proposed system focuses on filtering spam emails, categorizing messages based on importance, and estimating carbon emissions associated with email usage. This approach not only enhances efficiency in email management but also promotes environmentally responsible digital behavior.

Therefore, this study aims to highlight the hidden environmental cost of email communication and provide a practical, technology-driven solution to mitigate its impact, contributing to sustainable and eco-friendly computing practices.

2 Literature Review

The concept of the digital carbon footprint has gained significant attention with the rapid growth of digital technologies and online activities. Every digital action—including sending emails, streaming content, and using cloud services—consumes energy through devices, data transmission networks, and data centers, resulting in carbon emissions [5,6,9,22,23]. These emissions arise not only from usage but also from the production and disposal of electronic devices, making the digital carbon footprint a comprehensive measure of environmental impact [5]. Spatiotemporal studies have revealed significant variations in carbon emissions across different digital economy sectors, emphasizing the need for region-specific and sector-specific optimization strategies [23].

Email communication, one of the most widely used digital services, contributes notably to this footprint. Although individual emails appear environmentally insignificant, their cumulative impact is substantial. Studies show that standard emails generate approximately 0.3–4 g of CO_2, while emails with attachments can produce up to 50 g of CO_2 per email [4,8]. Spam emails, despite lower emissions per message, occur in large volumes and contribute significantly to overall emissions [1,4]. Research on spam detection using machine learning has demonstrated high accuracy rates in filtering unwanted emails, which can substantially reduce unnecessary data storage and transmission [19,20].

Recent research in green computing emphasizes the need for intelligent and sustainable digital communication systems. A study published in IEEE explores the environmental impact of digital communication and highlights the importance of optimizing resource usage and reducing unnecessary data transmission through efficient system design and management strategies [7]. The study suggests that integrating smart technologies can significantly improve energy efficiency in communication systems. Green data center initiatives have shown that implementing energy-efficient cooling systems, renewable energy sources, and intelligent workload management can reduce carbon emissions by up to 40% [18]. Furthermore, comprehensive surveys on email classification and filtering techniques have identified machine learning approaches as highly effective for intelligent email management [21].

Research also highlights that user behavior plays a critical role in emission reduction. For example, reducing just one unnecessary email per person per day can lead to significant national-level carbon emission reductions [2,10]. Additionally, lifestyle-based carbon footprint studies emphasize variations in emissions across different populations and sectors, using CO_2-equivalent (CO_2e) metrics for evaluation [3]. The integration of phishing detection systems and spam filters not only enhances security but also contributes to reducing the environmental impact of malicious email traffic [19].

However, many existing studies rely on estimated or generalized data and primarily focus on awareness and behavioral recommendations rather than practical implementation. While they successfully quantify emissions and identify key contributing factors, they lack intelligent, automated systems to manage and reduce digital carbon footprints [8,9]. Most research does not incorporate

real-time monitoring, predictive modeling, or machine learning techniques [4,6]. The global energy consumption of Information and Communication Technology (ICT) infrastructure continues to rise, with data centers accounting for a significant portion of this growth [22].

Furthermore, broader analyses of digital carbon footprints highlight the increasing energy demand of data centers and digital infrastructure but often fail to address specific optimization strategies for email communication [5,9,12,18]. There is also a lack of personalized emission tracking systems and scalable solutions applicable to diverse user populations [11,13].

Therefore, despite growing awareness of the environmental impact of digital communication, there remains a clear need for intelligent, AI-driven systems that can automatically optimize email usage, reduce unnecessary communication, and provide real-time carbon footprint estimation and analysis.

2.1 Gap Analysis

The gap analysis highlights the limitations and unresolved challenges in existing research related to digital carbon footprint and email communication. While prior studies provide valuable insights into emission estimation and awareness, they often lack intelligent, automated solutions for optimization. In particular, the absence of machine learning-based systems and real-time monitoring mechanisms remains a significant drawback. This analysis identifies key research gaps that motivate the development of an AI-driven, sustainable email management framework.

Table 1: Research Papers on Digital Carbon Footprint: Problems, Key Findings, and Identified Gaps

No.	**Problem**	**Key Findings**	**Identified Gaps**
[1]	Lack of automated systems for measuring email carbon footprint	Quantifies CO_2 emissions from spam, standard emails, and attachments.	No automation or machine learning-based optimization system.
[2]	Limited system support for behavioral email reduction strategies	Reducing email frequency can significantly lower carbon emissions.	Focuses only on user behavior; lacks AI-driven or system-level intervention.
[3]	Absence of digital/email-specific carbon footprint modeling	Provides CO_2e estimation and compares urban vs rural carbon footprints.	Does not address emissions from digital communication or emails.

No.	Problem	Key Findings	Identified Gaps
[4]	No intelligent optimization for email emissions reduction	Provides emission estimates and identifies key contributing factors.	Lacks AI/ML-based filtering or optimization mechanisms.
[5]	Lack of focused solutions for digital communication emissions	Explains environmental impact of digital activities and data centers.	No targeted solution for email-specific carbon reduction.
[6]	Absence of predictive systems for digital carbon footprint	Defines sources and environmental impacts of digital carbon emissions.	No technical or machine learning-based predictive model.
[7]	Lack of email-specific green computing applications	Highlights energy-efficient communication system design strategies.	Does not focus on email communication or real-time ML-based optimization.
[8]	No real-time email emission monitoring system	Classifies email types (spam, standard, attachments) and their emission levels.	No real-time tracking or automated filtering system.
[9]	Absence of automated tools for reducing email-related emissions	Reducing email usage leads to lower carbon footprint.	No automated tracking or optimization system.
[10]	Lack of personalized CO_2 emission estimation models	Provides general CO_2 emission calculation frameworks.	No personalization or predictive modeling (e.g., ML regression).
[11]	Email communication not specifically addressed in ICT energy studies	ICT systems contribute significantly to global energy consumption.	No focus on optimizing email communication specifically.
[12]	No implementation-level solution for digital sustainability	Discusses environmental impact of data growth and digital trends.	Lacks practical AI-based implementation for emission reduction.

2.2 Benchmark Analysis

To contextualize the proposed approach within the existing body of research, a comprehensive benchmark analysis was conducted comparing six significant studies in related domains. This analysis evaluates various dimensions including

Table 2. Benchmark analysis of related works.

Feature	**GEECO [18]**	**Phishing Survey [19]**	**Spam Study [20]**	**Email Class. Review [21]**	**ICT Carbon [22]**	**Our Paper**
Problem Focus	Data center energy + carbon	Phishing detection	Spam emails	Email classification	ICT carbon emissions	Email carbon + optimization
Scope	Medium–Broad	Narrow	Narrow	Medium–Broad	Very broad	Focused
Approach	Implementation + modeling	Survey	Analytical	Survey	Empirical + statistical	Implementation + ML
ML Usage	Moderate	Reviewed	No	Reviewed	No	Strong
Dataset	Simulated / estimated	Not specified	No	Strong datasets	Large real datasets	Limited / estimated
System Implementation	Yes	No	No	No	No	Yes
Sustainability Focus	Strong	No	No	No	Strong	Strong

problem focus, scope, methodological approach, use of machine learning, dataset quality, system implementation, and sustainability focus. The comparative analysis demonstrates how this work builds upon and extends previous research by combining strong machine learning implementation with practical system deployment and a focused sustainability perspective.

3 Proposed Methodology

This section presents a systematic, data-driven methodology for reducing the digital carbon footprint of email communication using a multi-model machine learning approach. Unlike traditional approaches that rely solely on user awareness or static estimations, the proposed method integrates intelligent automation through spam detection, importance classification, and carbon emission estimation. The methodology is designed to be practical, scalable, and applicable in real-world digital communication systems.

3.1 Overview of the Proposed Framework

The proposed methodology follows a structured pipeline consisting of dataset utilization, data preprocessing, spam detection, importance classification, carbon emission estimation, and system-level evaluation. The framework is designed as a sequential decision-making process where each stage refines the output of the previous stage.

The overall workflow of the system is as follows:

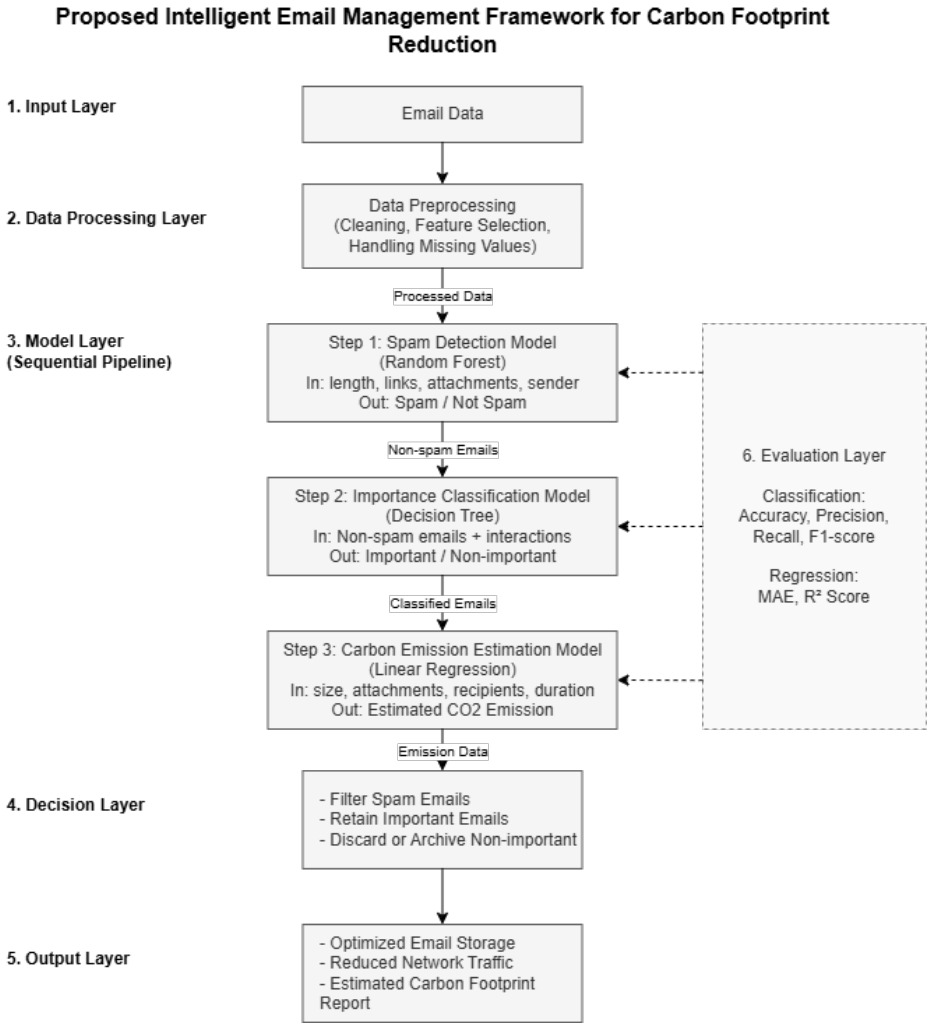

Fig. 1. Proposed Intelligent Email Management Framework for Carbon Footprint Reduction

This pipeline ensures that unnecessary emails are filtered at an early stage, thereby reducing storage, processing requirements, and associated energy consumption.

3.2 Data Source and Preparation

Two separate datasets are used in this study to ensure task-specific model training.

The *combined_ data.csv* dataset is used for training the spam detection and importance classification models. It contains features such as email length, number of links, attachment presence, sender type, subject characteristics, and interaction frequency. The dataset also includes target labels *is_ spam* and *importance*.

The *email_ carbon_ dataset.csv* dataset is used for training the carbon emission estimation model. It includes features such as email size, attachment size, number of recipients, and storage duration, along with the target variable *carbon_ emission*.

Data preprocessing involves cleaning missing values, removing inconsistencies, and selecting relevant features for each model. This ensures improved model performance and reliability.

3.3 Spam Detection Model

The first stage of the framework focuses on identifying and filtering spam emails using a Random Forest classifier. The model is trained on features such as email length, number of links, attachment presence, sender type, and subject length.

The target variable *is_ spam* is defined as:

- Level 1: Spam email
- Level 0: Non-spam (legitimate) email

This stage plays a critical role in reducing unnecessary email traffic, which directly contributes to minimizing storage requirements and energy consumption.

3.4 Importance Classification Model

The second stage classifies emails based on their importance using a Decision Tree classifier. This model uses features such as sender type, subject-related attributes, and frequency of interaction. Additionally, the predicted output from the spam detection model is incorporated as an input feature to improve classification performance.

Only non-spam emails are processed in this stage. Emails are categorized into important and non-important classes, enabling efficient prioritization and reducing unnecessary data retention.

3.5 Carbon Emission Estimation

The environmental impact of email communication is estimated using a Linear Regression model trained on the *email_ carbon_ dataset.csv* dataset. The model

predicts carbon emissions based on input features such as email size, attachment size, number of recipients, and storage duration.

The estimation is based on the standard linear regression relationship:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \tag{1}$$

Where, y represents carbon emission, x_1, x_2, x_3 represent input features, and β represents the learned coefficients.

This model provides a quantitative estimation of the carbon footprint associated with each email.

3.6 Scenario-Based Analysis

To evaluate the effectiveness of the proposed system, different operational scenarios are considered:

- **Baseline Scenario:** All emails are processed and stored without filtering.
- **Optimized Scenario:** Spam emails are removed and only important emails are retained.
- **Inefficient Scenario:** All emails, including spam and non-important ones, are stored.

This comparison helps analyze how intelligent email management can reduce energy consumption and carbon emissions.

3.7 Validation and Sustainability Perspective

The proposed system is evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score for classification models, and Mean Absolute Error (MAE) and R^2 score for the regression model.

Although the carbon emission values are estimated rather than directly measured, they provide a reasonable approximation of environmental impact based on email characteristics. From a green computing perspective, the proposed approach offers a practical and low-cost solution that does not require additional hardware or sensor-based infrastructure.

Acknowledgments. The authors would like to acknowledge the support of the Department of Computer Science and Engineering, United International University, Dhaka, Bangladesh.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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